Submission Summary

Conference Name

2nd International Conference on Electronics, Computing, Communication and Networking Automation Technologies 2026

Paper ID

3949

Paper Title

A Smart Computing Framework for AI-Driven Urban Drainage and Waterlogging Prediction

Abstract

A multi-source Artificial Intelligence (AI) framework for enhancing and maintaining urban waterlogging prediction is presented in this paper. It achieves this through the combination of actual weather data, drainage maps created using a Geographic Information System (GIS), and Internet of Things (IoT) water level sensors, providing highly accurate and current flood information. The increasing issues of urban flooding, made worse by climate change, rapid urban growth, and poor drainage systems, are the main reasons for this study. To better understand how drainage works, the proposed framework combines information from IoT-based water level networks, spatial drainage models, and data from local weather stations. Using AI methods such as ensemble learning and Long Short Term Memory (LSTM) neural networks, the system analyzes both live and historical data to make reliable and data-driven predictions of incidents of possible waterlogging. Integration of several data sources, dynamic updating of risk profiles through real-time sensor feedback, and the combination of data-driven AI models and hydrodynamic simulations are what make it novel. The framework's effectiveness is demonstrated through structured simulation-based experimental validation. Overall, this framework advances the development of intelligent and climate resilient urban drainage systems, supporting the more general objectives of adaptive and sustainable urban development. Index Terms—GIS-based drainage mapping, IoT water-level sensing, LSTM neural networks, ensemble learning, real-time monitoring, climate-resilient infrastructure, smart cities, artificial intelligence (AI), multi-source data fusion, sustainable drainage systems, waterlogging prediction, urban flood management.

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Submission Files

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